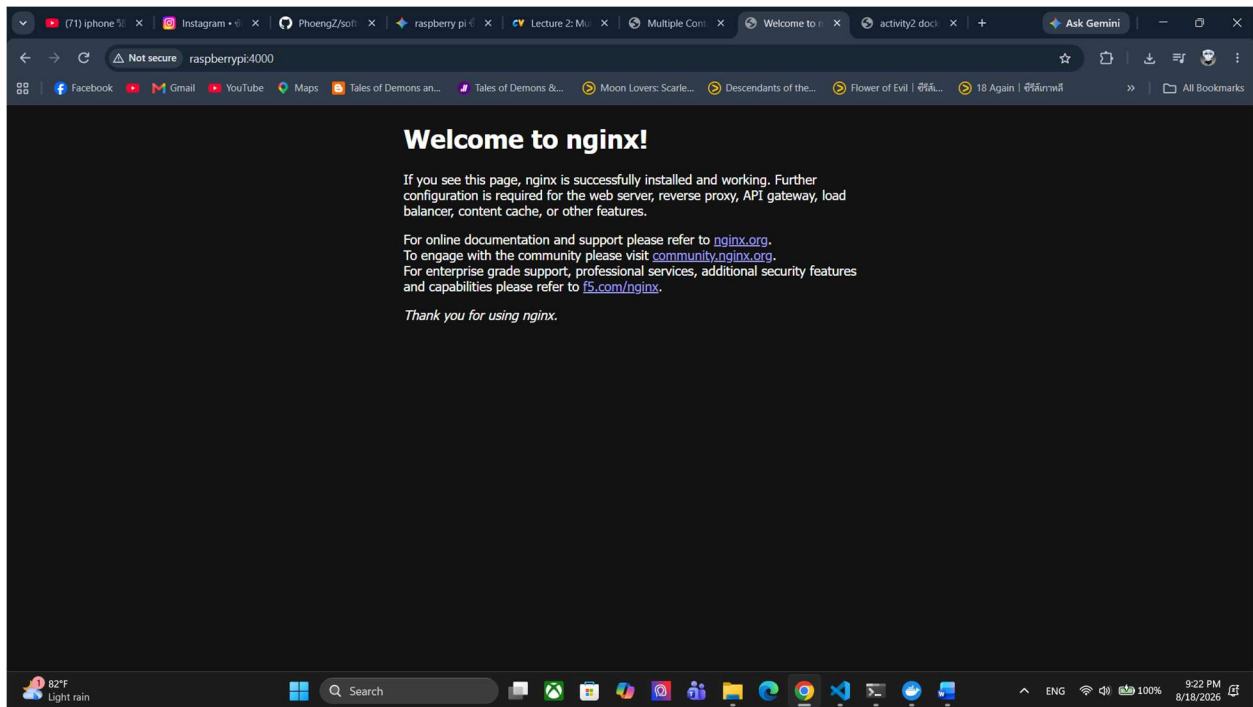
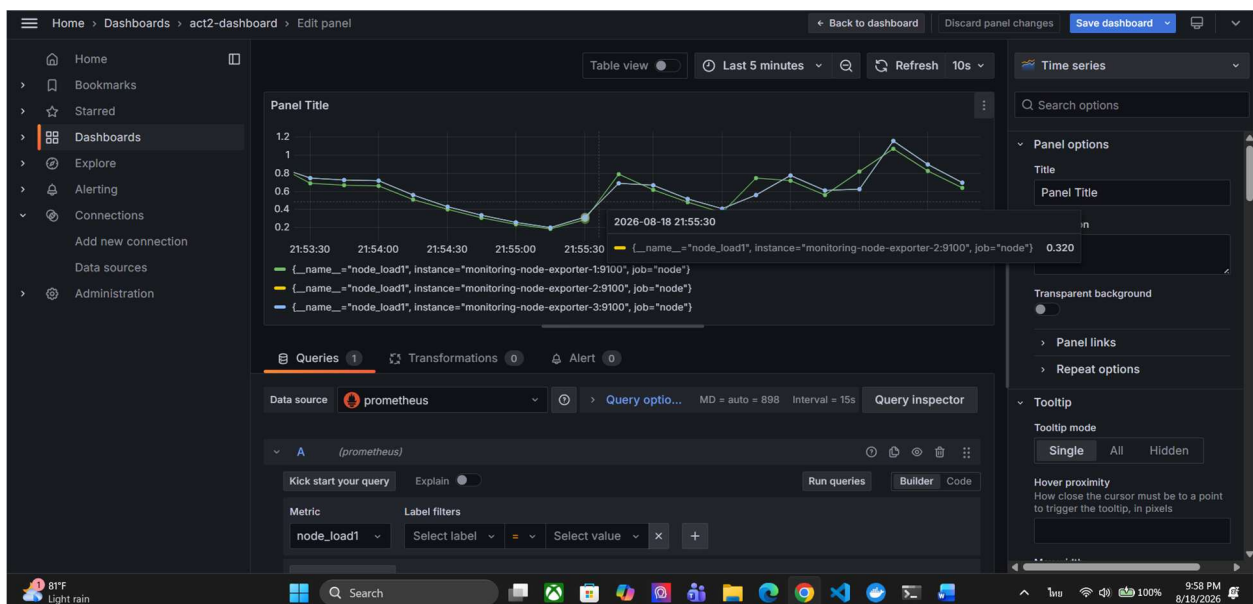




Part 1: Docker compose บน computer เครื่องเดียว

CONTAINER ID	NAME	CPU %	MEM USAGE / LIMIT	MEM %	NET I/O	BLOCK I/O	PIDS
ca3f1bb7cf61	prometheus	0.43%	38.55MiB / 7.366GiB	0.51%	1MB / 101kB	0B / 0B	18
7b1670bdee15	grafana	0.03%	99.29MiB / 7.366GiB	1.32%	16.2MB / 17.2MB	0B / 0B	24
772f84d6a2d9	monitoring-node-exporter-3	26.62%	13.1MiB / 7.366GiB	0.17%	19.4kB / 312kB	0B / 0B	5
40e071efafd3	monitoring-node-exporter-1	25.29%	14.74MiB / 7.366GiB	0.20%	19.1kB / 311kB	0B / 0B	6
da8435edfd72	monitoring-node-exporter-2	26.04%	15.23MiB / 7.366GiB	0.20%	19kB / 311kB	0B / 0B	6
87ec29350c1c	pg_container	0.00%	21.52MiB / 7.366GiB	0.29%	1.95kB / 126B	0B / 0B	7
644bfd7c1b57	pgadmin4_container	0.03%	188MiB / 7.366GiB	2.49%	1.95kB / 126B	0B / 0B	6



```
(base) PS C:\Users\USER\Desktop\Software_def\act2> .\activity2-CP_windows_amd64.exe
2026/08/24 17:33:09 Current Timestamp: 2026-08-24T17:33:09+07:00
2026/08/24 17:33:09 ✓ / Container grafana exists.
2026/08/24 17:33:09 ✓ / Container prometheus exists.
2026/08/24 17:33:09 ✓ / Container monitoring-node-exporter-1 exists.
2026/08/24 17:33:09 ✓ / Container monitoring-node-exporter-2 exists.
2026/08/24 17:33:09 ✓ / Container monitoring-node-exporter-3 exists.
2026/08/24 17:33:09 ✓ / All specified containers are running.
2026/08/24 17:33:09 ✓ / Docker compose is running.
2026/08/24 17:33:09 ✓ / Network exists.
2026/08/24 17:33:10 ✓ / URL http://localhost:3000 returns 200.
2026/08/24 17:33:10 ✓ / Grafana is up and running at http://localhost:3000
2026/08/24 17:33:10 ✓ / URL http://localhost:9090 returns 200.
2026/08/24 17:33:10 ✓ / Prometheus is up and running at http://localhost:9090
2026/08/24 17:33:10 Result: true
2026/08/24 17:33:10 🎉 Looks good! Please enter your StudentID and Full name below
👉 StudentID: 6630199021
👉 Full Name (TH): เผ่าลาภ กุลธีระ
2026/08/24 17:33:26 🎉 Successfully submitted. Your lucky number is 21381054485523375
2026/08/24 17:33:26 🎉 Congratulations! You have completed the activity 🎉🎉🎉
2026/08/24 17:33:26 ⚠️ Don't forget! you still need to submit your assignment via MyCourseVille ⚠️
(base) PS C:\Users\USER\Desktop\Software_def\act2> |
```

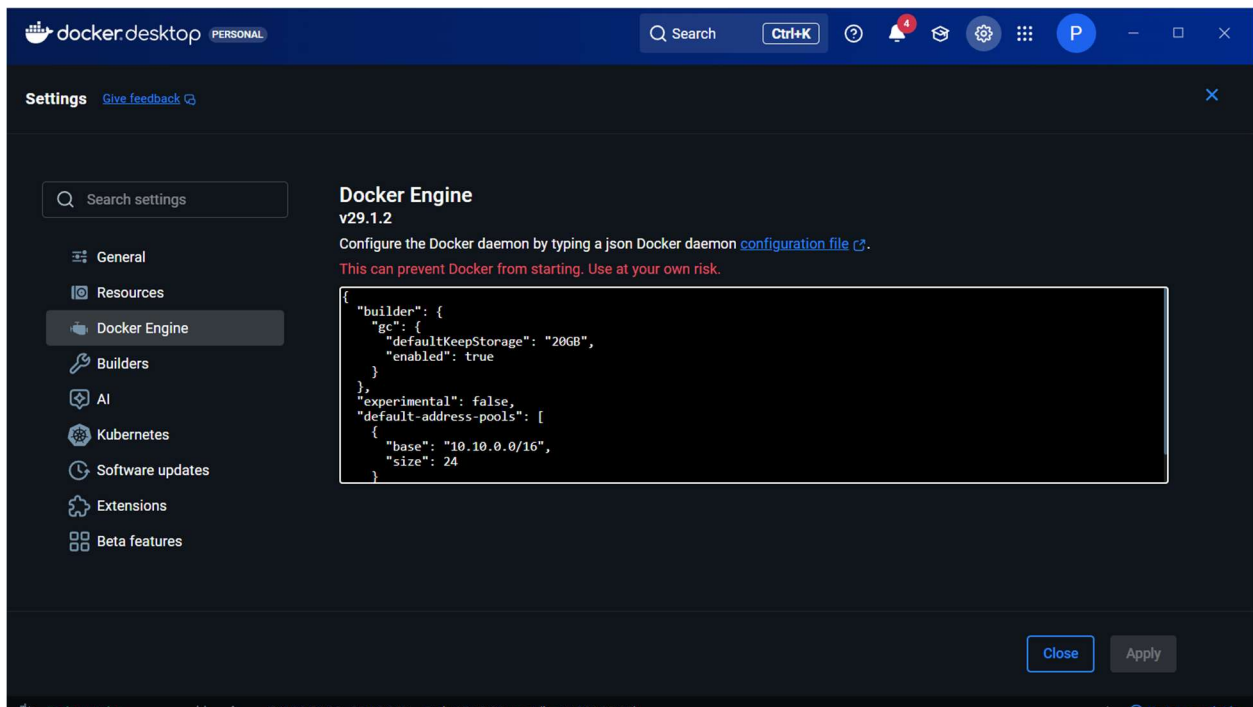
Part 2: Docker

เกิดปัญหา เนื่องจาก host pc สามารถเปิดดู metrics ได้ผ่าน Web แต่ว่า Prometheus ไม่สามารถ scrape มาได้โดยการทำ Tcpdump ที่ port 9100 ของ node-exporter บน raspberrypi

```
pi@raspberrypi:~$ sudo tcpdump -i wlan0 port 9100 -nn
tcpdump: verbose output suppressed, use -v[v]... for full protocol decode
listening on wlan0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
14:54:43.452173 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 3993105512, win 501, length 0
14:54:43.463533 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:54:58.212139 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], seq 0:1, ack 1, win 255, length 1
14:54:58.212439 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, options [nop,nop,sack 1 {0:1}], length 6
14:55:13.404188 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:55:13.415674 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:55:28.508171 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:55:28.518869 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:55:43.267833 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], seq 0:1, ack 1, win 255, length 1
14:55:43.268166 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, options [nop,nop,sack 1 {0:1}], length 6
14:55:58.460181 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:55:58.471410 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:56:13.564171 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:56:13.581322 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:56:28.323459 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], seq 0:1, ack 1, win 255, length 1
14:56:28.323759 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, options [nop,nop,sack 1 {0:1}], length 6
14:56:43.516250 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:56:43.527720 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:56:58.620175 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:56:58.632982 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:57:13.380126 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], seq 0:1, ack 1, win 255, length 1
14:57:13.380425 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, options [nop,nop,sack 1 {0:1}], length 6
14:57:28.572182 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:57:28.583539 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:57:43.676171 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:57:43.689280 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:57:58.434929 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], seq 0:1, ack 1, win 255, length 1
14:57:58.435244 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, options [nop,nop,sack 1 {0:1}], length 6
14:58:13.628187 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:58:13.641684 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:58:28.732177 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
14:58:28.744587 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], ack 1, win 255, length 0
14:58:43.490759 IP 172.20.10.2.54715 > 172.20.10.4.9100: Flags [..], seq 0:1, ack 1, win 255, length 1
14:58:43.491062 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, options [nop,nop,sack 1 {0:1}], length 6
14:58:58.604133 IP 172.20.10.4.9100 > 172.20.10.2.54715: Flags [..], ack 1, win 501, length 0
```

จะเห็นว่า ไม่มีการส่ง syn ack หรือส่ง request http เลยแม้จะ restart Prometheus ก็ตาม เลยคิดว่า เลยคิดว่า ปัญหา น่าจะเกิดจากการที่ docker เติค route table แล้วคิดว่าเป็น ip ใน docker อีกทั้ง hostpot ดันแจก IP เดียวกันกับที่ docker แจกด้วยทำให้ มันเกิดการสับสน ทำให้ docker เลยไม่ส่ง request ออกมายัง default gateway ทำให้ request มาไม่ถึง

เลยทำการแก้ไขการแจก IP ของ docker ที่ docker engine โดยกำหนดให้ แจก private IP



จากนั้น docker compose up -d ใหม่ก็จะขึ้น node-exporter-2 บน raspberrypi เรียบร้อย

